Luke Parrella

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Skills

Languages:	Python, NodeJS, C, MATLAB
AI/ML:	Pydantic AI Agents, LLM Integration, Text-to-SQL
Technology:	Microcontrollers, Radio Communication, Observability with Prometheus and Grafana
Tools:	SQLite, Ansible, CI/CD

Projects

BO-AT Autonomous sailing vessel — August 2025 - Present

- Collaborated on a 7-person team to architect an autonomous ocean vessel system capable of carrying modular payloads, contributing to physical control and communication subsystems
- Designed and iterated on a rudder control mechanism using onshape, coordinating with the software team to ensure mechanical characteristics met the requirements of the autonomous control system
- Communicated with faculty mentor weekly and the Woods Hole Oceanographic Institute, to align the project with client objectives and ensure market viability

Wildlife ID — April 2026

- Built a wildlife identification pipeline using a decoupled pub/sub architecture, enabling independent scaling of ingestion, inference, and storage components
- Designed a dual-database strategy combining a document store for image annotations and a vector DB for embedding-based similarity search, optimizing query latency
- Maintained high code quality through CI/CD pipelines and unit testing, achieving reliable deployment across development cycles

DataViz - Snowleopard Hackathon, *third place* — March 2026

- Built a pydantic-ai agent that autonomously selects visualization types from Snowleopard query results and generates Grafana dashboard panels via API, delivered within a 3-hour hackathon time constraint
- Leveraged Claude Code to accelerate prototyping, demonstrating ability to ship functional demos rapidly under competitive conditions

SpeedySheets — March 2026

- Built a Python CLI application using Click that ingests CSV files, dynamically infers schema from column headers and pandas dtypes, and loads data into a SQLite database via a custom ETL pipeline
- Implemented a text-to-SQL engine that injects live schema context into LLM prompts, parses the generated SQL and executes it against the database - enabling plain-English queries on large datasets

Minecraft Server Status and Display — February 2026

- Integrated public APIs to continuously poll server and player telemetry, building a live status display pipeline
- Automated deployment using Ansible playbooks, reducing environment setup to a single command and enabling reproducible infrastructure

Work Experience

HITA Energy — Boston, MA
Founding Engineer — October 2024 - June 2025

- Served as a founding engineer at an angel-backed startup, collaborating across a cross-disciplinary team of software and materials engineers
- Designed and implemented control system in Python that interfaced with real hardware including pneumatic controls and bitcoin ASIC mining units
- Built a relay-based hardware control system containing relays to control high and low voltage power in order to command and receive data from pumps, valves, and thermistors

Education

Boston University — Boston, MA
Bachelor of Science Computer Engineering — Expected Graduation May 2026